



WEEK 3 · PROJECT 1

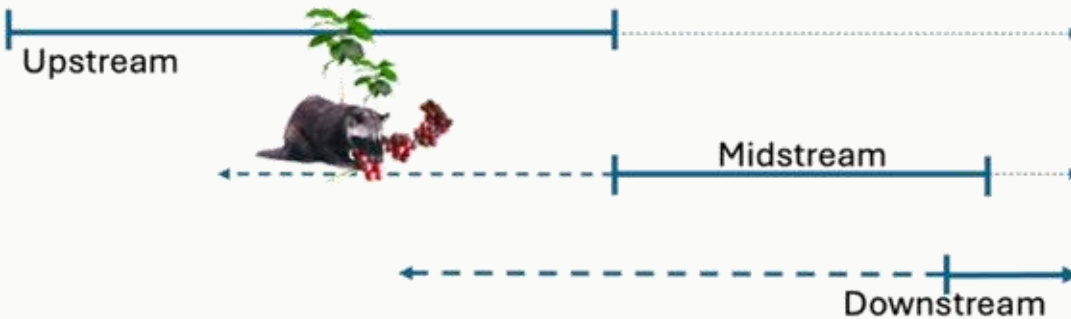
COFFEE LAND vs PRICE

EDA & SQL PROJECT PRESENTATION
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2 OCTOBER 2026

THE COFFEE CHAIN

FROM FARM TO CUP



What drives coffee's export price?

- Do bigger, more efficient growers sell cheaper?
- Do buyers pay more than sellers actually get?

THREE OFFICIAL RECORDS :

GREEN COFFEE PRODUCTION, PROCESSED GREEN BEANS, AND TRADE

64

years

1961-2024

203

countries

represented in the coffee chain

41,179

official values

8,082 production rows

33,097 trade rows

SOURCE: FAOSTAT ANALYTICAL DATA.

SOURCE DOWNLOAD

DATA QUALITY

IGNORE

Estimated or unverified values

Inputs that do not meet the verification rule.

KEEP

Officially verified values

Inputs that remain comparable across the analysis.

RESULT

Processed dataset

Two usable data records inputs remained after filtering.

DATA MODEL

HOW THE COFFEE DATA CONNECTS

Reference tables hold countries, measurements, and data-quality flags.

Record tables hold production and import/export price observations.

Linked keys connect each observation to its country, year, measurement, and source.

Coffee database (FAOSTAT, Coffee, green) — ERD

Lookup tables (parents)

countries		203 rows
PK	area_code	INTEGER
	area_name	TEXT

elements		7 rows
PK	element_code	INTEGER
	element_name	TEXT
	unit	TEXT

flags		1 row
PK	flag	TEXT
	flag_description	TEXT

Fact tables (children)

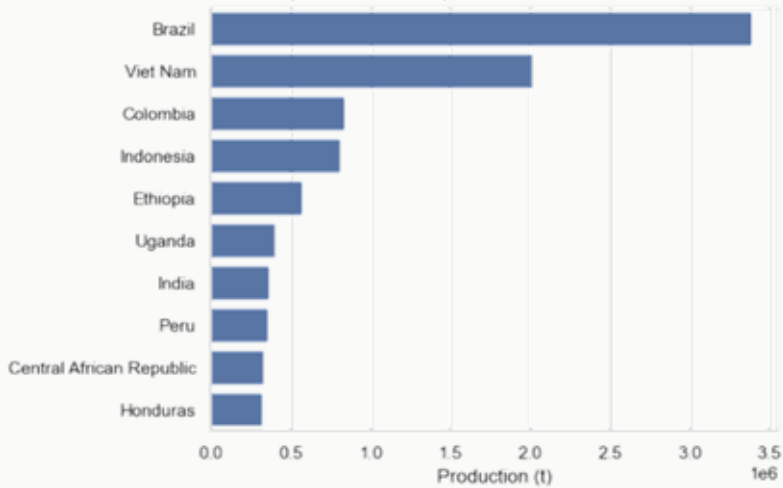
production		8,082 rows
PK, FK	area_code	INTEGER
PK	year	INTEGER
PK, FK	element_code	INTEGER
	value	REAL
FK	flag	TEXT

trade		33,097 rows
PK, FK	area_code	INTEGER
PK	year	INTEGER
PK, FK	element_code	INTEGER
	value	REAL
FK	flag	TEXT

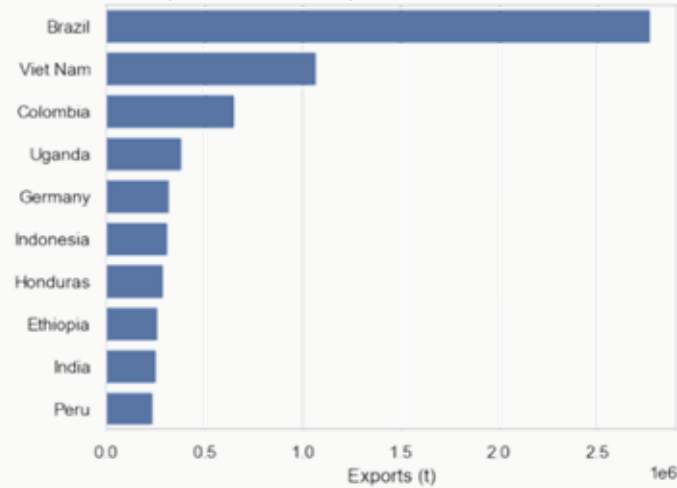
ENTITY RELATIONAL DIAGRAM (ERD)
SOURCE: SUPPLIED POWERPOINT CONTENT

FINDINGS

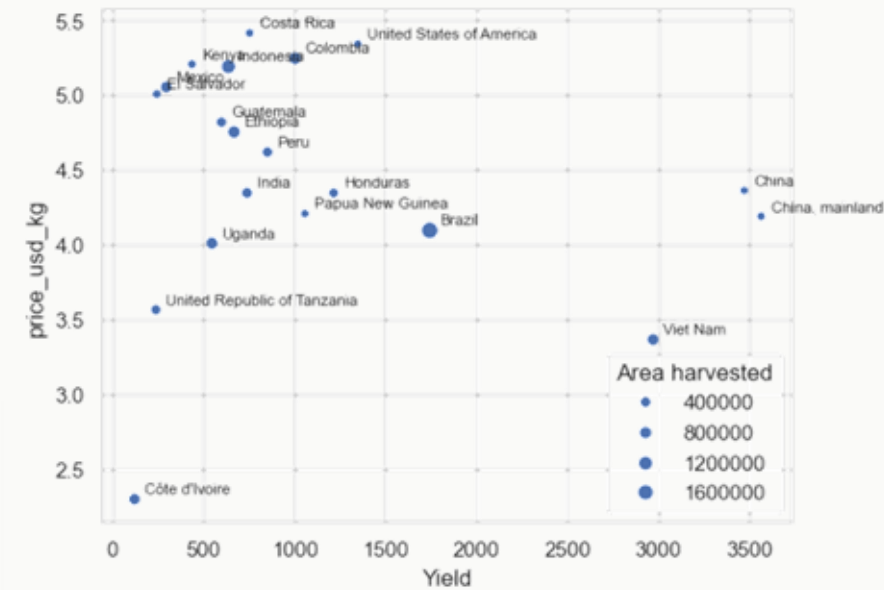
Top 10 coffee producers, 2024 (tonnes)



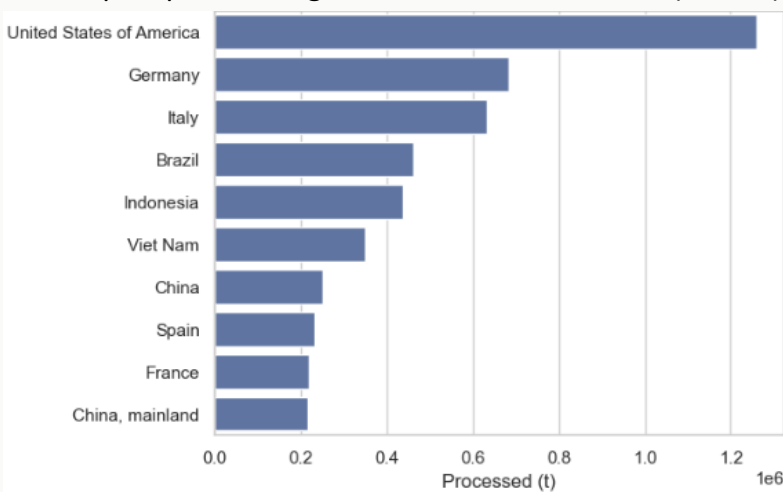
Top 10 coffee exporters, 2024 (tonnes)



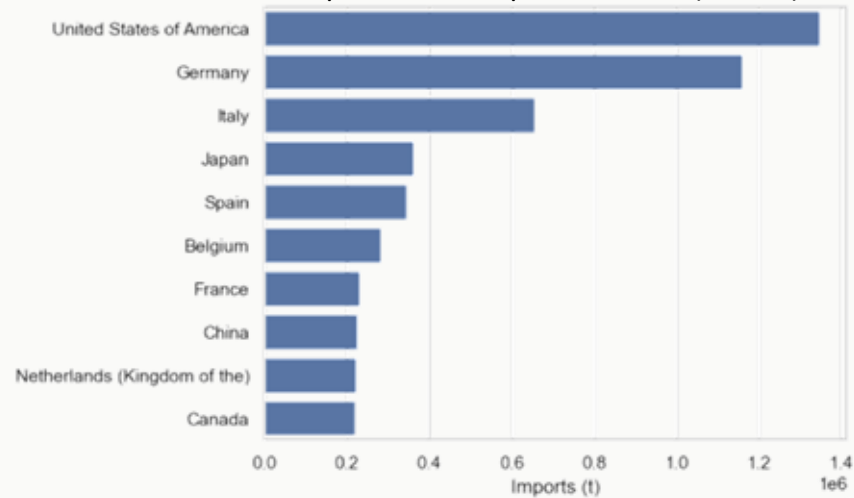
Yield vs export price, 2024 (tonnes)



Top 10 processed-green beans volume , 2024 (tonnes)



Top 10 coffee importers, 2024 (tonnes)



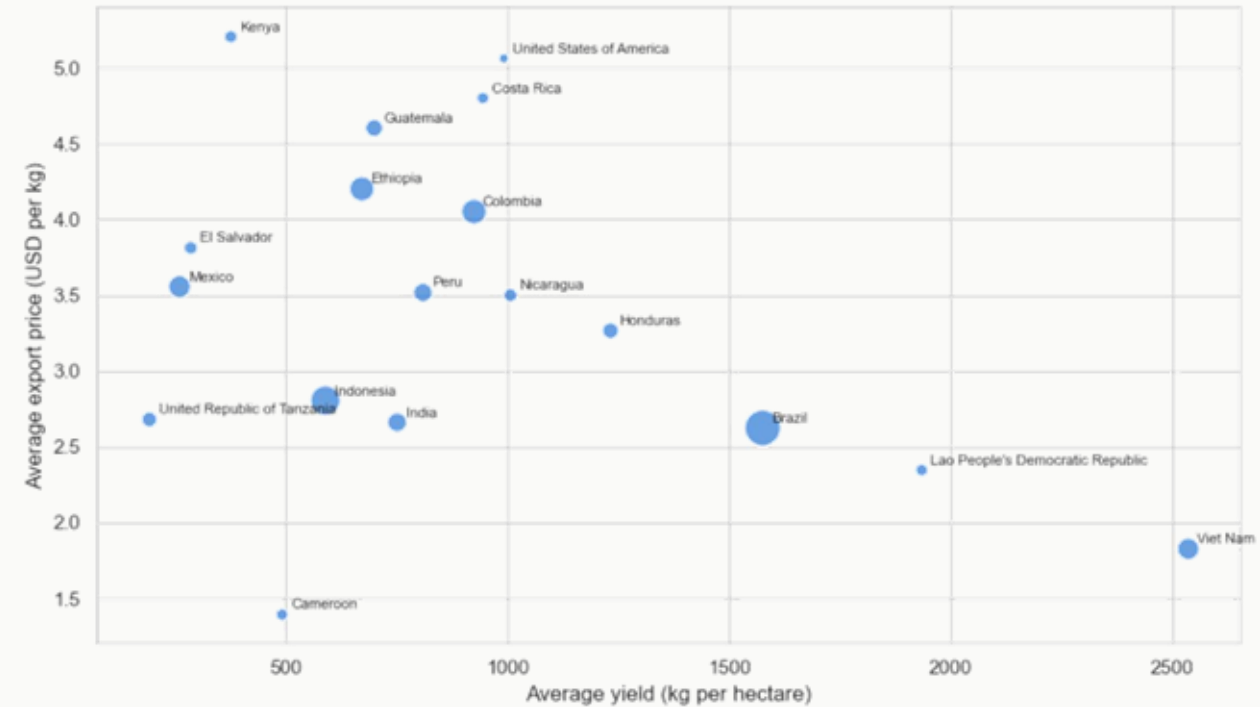
CONCLUSIONS

Do more land or higher yield mean a lower export price?

Weakly supported. Larger, higher-yield producers tend to sell a little cheaper, but the relationship is weak and inconsistent across all 18 countries.

Area correlation: **-0.16** · Yield correlation: **-0.26**

Viet Nam: highest yield **2,536 kg/ha**, cheapest price **\$1.83/kg**. Kenya: lowest yield **376 kg/ha**, highest price **\$5.21/kg**.



Do importers pay more per kg than exporters receive?

Partly supported. Importers paid more in **54 of 64 years (84%)**, but the gap did not grow.

The gap peaked around 1990 at **+27%** and nearly disappeared by 2024 at **-0.5%**.

- FAIR TRADE
- DIRECT TRADE
- FARM TO TABLE



WHATS NEXT

- Arabica vs. robusta is not distinguished in the data.
- Build a tech-access index and re-run the yield–price test against the total yield of coffee farmers with ‘better tech’.

T H A N K Y O U .

